CYBERSECURITY | INFORMATION SECURITY

Cybersecurity Incident Intelligence & Threat Analytics Dashboard

SQL and Power BI analysis of the EuRepoC Global Dataset v1.3



EuRepoC Global Dataset v1.3

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Project Overview

The problem

Security teams need to understand historical incident patterns, threat categories, affected receiver groups, geographic distribution and incident intensity. Public incident data is hard to explore consistently without structured querying and visual summaries.

The solution

MySQL for storage and SQL profiling, then Power Query, DAX and Power BI visuals to summarize the dataset in an interactive, filterable dashboard.



INDUSTRY

Cybersecurity / Information Security



DATASET

EuRepoC Global Dataset v1.3



DATABASE

cybersecurity_incident_db

Project Objectives

1. Analyze documented cybersecurity incidents using SQL and Power BI
2. Examine incident trends across the available years
3. Explore incident types and initiator / receiver categories
4. Review weighted intensity and flag higher-intensity records (≥ 7)
5. Deliver an interactive dashboard with filtering and exploration

Dataset at a Glance

3,414

incident records imported

3,414

distinct incident IDs

2000–2024

1 Jan 2000 to 19 Dec 2024

3,320

economic_impact unavailable or missing (missing is not zero)

17

receiver_category values missing

Project Workflow

1. **Industry Selection**
   Cybersecurity incident intelligence
2. **Problem Definition**
   Trends, categories, geography, intensity
3. **Dataset Collection**
   EuRepoC v1.3 workbook
4. **MySQL Setup**
   Database and raw table import
5. **Data Cleaning**
   SQL view and Power Query dates
6. **SQL Analysis**
   Counts, missing values, date range
7. **Power BI Modeling**
   MySQL connection and DAX
8. **Dashboard Build**
   Three pages with year slicer
9. **Insights**
   Findings and limitations

Tools & Technologies

MySQL Server 8.0.46
Structured storage of incident records

SQL
Profiling, validation and analysis

Power BI Desktop
Interactive dashboard and visuals

Power Query
Date parsing and field preparation

DAX
Measures and calculated columns

Excel (.xlsx)
Source dataset and import workflow

SQL Analysis & Cleaning

Checks performed

- Row counts and distinct incident-ID checks
- Missing-value profiling on key fields
- Date parsing and range checks (dd.MM.yyyy)
- weighted_intensity frequency checks
- Clean view with standardized receiver country

```sql
-- Weighted intensity distribution
SELECT weighted_intensity,
       COUNT(*) AS incident_count
FROM eurepoc_incidents_raw
GROUP BY weighted_intensity
ORDER BY weighted_intensity;

-- Unique incident IDs
SELECT COUNT(DISTINCT incident_id)
FROM eurepoc_incidents_raw;
```

eurepoc_incidents_clean: blank, null and “Not available” receiver countries become “Unknown”. The raw table is preserved.

Power BI Data Model

Data preparation

- Connected to MySQL view eurepoc_incidents_clean
- Parsed start_date into start_date_clean (dd.MM.yyyy); original kept
- Intensity fields set as whole numbers
- economic_impact kept as text; missing values not set to zero

DAX measures

- **Incident Count**
- **Years Covered**
- **Max Intensity**
- **Initiator Categories**
- **Countries**
- **High Intensity Incidents**
- **Most Common Initiator**
- **Most Common Receiver**
- **Multi-Type Incidents**

Calculated column: Incident Year (for time-based analysis)

Dashboard Structure

Executive Overview

- Total incident count
- Years covered
- Max weighted intensity
- Incidents by year
- Top incident types

Threat Analysis

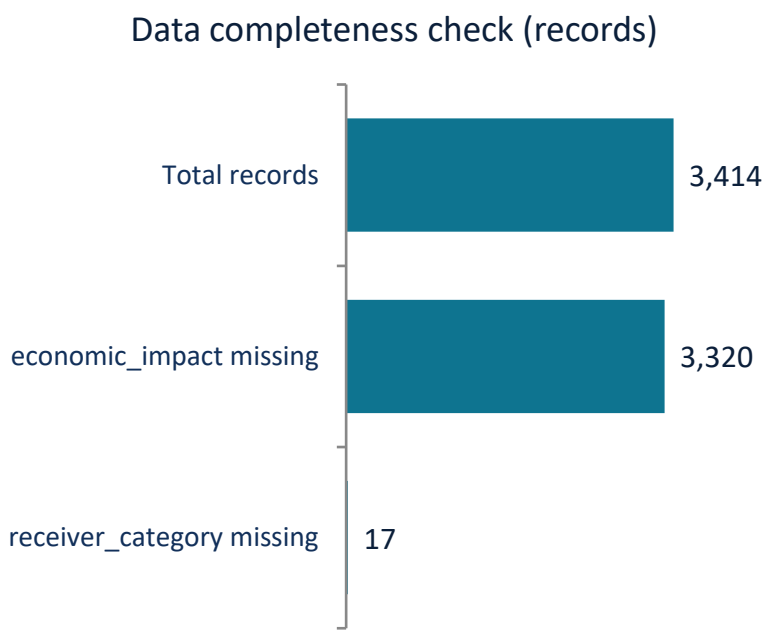
- High-intensity incident count
- Most common initiator
- Most common receiver
- Multi-type incidents
- Intensity distribution

Geography & Impact

- Receiver-country distribution
- Known-country count
- Impact-related fields

A year slicer filters every page. High intensity means weighted_intensity ≥ 7 (project-defined). Dark navy and cyan theme.

Key Insights from the Data



- **3,414 unique records.** Row count equals distinct incident IDs.
- **Long time span.** Dates run from 1 Jan 2000 to 19 Dec 2024.
- **Sparse economic impact.** Unavailable for 3,320 records; missing is not zero.
- **Receiver category gaps.** 17 records have no value.
- **Intensity scale.** Weighted intensity ranges from 0 to 11.
- **Multi-value types.** Semicolon-separated values mean distinct raw values are not individual types.

Limitations

Historical dataset, not a live security-event feed

Coverage and recording practices can affect trends

Several fields have substantial missing data

Semicolon-separated categories are not normalized

The ≥ 7 intensity threshold is project-defined, not an official standard

No response-time, MTTA/MTTR, analyst or patch metrics

Thank You

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Structured SQL profiling of 3,414 incident records



Interactive Power BI dashboard with year filtering



Transparent documentation of data-quality limits

Dataset: European Repository of Cyber Incidents (EuRepoC), Global Dataset v1.3. Educational and portfolio project; please follow the dataset's licensing and usage terms.